

hw1

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This is the link to the repo: <https://github.com/YifengHuang-Bob/Homework1>

Problem 1 - Exploratory Analysis of Soybean Cultivars

a.

```
soybean <- read.csv("data/data.csv")

names(soybean) <- c(
  "Season", "Cultivar", "Repetition", "PH", "IFP", "NLP",
  "NGP", "NGL", "NS", "MHG", "GY"
)

class(soybean)
```

```
[1] "data.frame"
```

```
names(soybean)
```

```
[1] "Season"      "Cultivar"    "Repetition"  "PH"          "IFP"
[6] "NLP"         "NGP"        "NGL"        "NS"          "MHG"
[11] "GY"
```

b.

```
nrow(soybean)
```

```
[1] 320
```

```
length(unique(soybean$Cultivar))
```

```
[1] 40
```

```
length(unique(soybean$Season))
```

```
[1] 2
```

```
table(soybean$Season)
```

```
 1  2  
160 160
```

The dataset contains 320 observations, 40 distinct cultivars, and observations from 2 seasons. There are 160 observations from Season 1 and 160 observations from Season 2.

c.

1.

```
cor(soybean$PH, soybean$GY)
```

```
[1] 0.1232807
```

The correlation between plant height (PH) and grain yield (GY) is approximately 0.123.

2.

```
cor(  
  soybean$PH[soybean$Season == 1],  
  soybean$GY[soybean$Season == 1]  
)
```

```
[1] 0.2308581
```

```
cor(  
  soybean$PH[soybean$Season == 2],  
  soybean$GY[soybean$Season == 2]  
)
```

```
[1] -0.0227046
```

The correlation between PH and GY is approximately 0.231 for Season 1 and -0.023 for Season 2. Therefore, Season 1 has the higher correlation.

3.

```
soybean$GY[which.max(soybean$PH)]
```

```
[1] 3304.678
```

The grain yield of the soybean observation with the greatest plant height is 3304.678.

4.

```
mean_mhg <- mean(soybean$MHG)
```

```
mean_mhg
```

```
[1] 168.3223
```

```
mean(soybean$MHG > mean_mhg) * 100
```

```
[1] 47.1875
```

47.1875% of the observations have an MHG greater than the overall average MHG.

5.

```
mean_gy_by_cultivar <- tapply(
  soybean$GY,
  soybean$Cultivar,
  mean
)

mean_gy_by_cultivar[which.max(mean_gy_by_cultivar)]
```

```
98R30 CE
3964.968
```

The cultivar with the highest average grain yield is 98R30 CE, with an average grain yield of 3964.968.

d.

```
quantitative_vars <- c(
  "PH", "IFP", "NLP", "NGP", "NGL", "NS", "MHG", "GY"
)

average_table <- rbind(
  Overall = colMeans(soybean[quantitative_vars]),
  `Season 1` = colMeans(
    soybean[soybean$Season == 1, quantitative_vars]
  ),
  `Season 2` = colMeans(
    soybean[soybean$Season == 2, quantitative_vars]
  )
)

round(average_table, 2)
```

	PH	IFP	NLP	NGP	NGL	NS	MHG	GY
Overall	68.39	15.46	59.09	135.09	2.29	4.07	168.32	3418.55
Season 1	65.99	16.48	57.97	135.24	2.30	4.82	162.29	3428.45
Season 2	70.79	14.45	60.21	134.94	2.28	3.33	174.36	3408.66

e.

1.

Let μ_1 be the mean grain yield (GY) for Season 1 and μ_2 be the mean grain yield for Season 2.

$$H_0 : \mu_1 = \mu_2$$

$$H_A : \mu_1 \neq \mu_2$$

2.

```
t.test(
  soybean$GY[soybean$Season == 1],
  soybean$GY[soybean$Season == 2]
)
```

Welch Two Sample t-test

data: soybean\$GY[soybean\$Season == 1] and soybean\$GY[soybean\$Season == 2]

```
t = 0.35155, df = 231.85, p-value = 0.7255
alternative hypothesis: true difference in means is not equal to 0
95 percent confidence interval:
 -91.15712 130.75170
sample estimates:
mean of x mean of y
 3428.452  3408.655
```

Manual calculation

```
gy1 <- soybean$GY[soybean$Season == 1]
gy2 <- soybean$GY[soybean$Season == 2]

n1 <- length(gy1)
n2 <- length(gy2)

mean1 <- mean(gy1)
mean2 <- mean(gy2)

var1 <- var(gy1)
var2 <- var(gy2)

t_stat <- (mean1 - mean2) /
  sqrt(var1 / n1 + var2 / n2)

df <- (var1 / n1 + var2 / n2)^2 /
  ((var1 / n1)^2 / (n1 - 1) +
   (var2 / n2)^2 / (n2 - 1))

p_value <- 2 * pt(-abs(t_stat), df = df)

t_stat
```

```
[1] 0.351546
```

```
df
```

```
[1] 231.8466
```

```
p_value
```

```
[1] 0.725498
```

The mean grain yield is 3428.452 for Season 1 and 3408.655 for Season 2. The Welch two-sample t-test gives a p-value of 0.7255. Since the p-value is greater than 0.05, we fail to reject the null hypothesis. There is insufficient evidence to conclude that the mean grain yield differs between the two seasons.

Problem 2 - Adult Income Dataset

a.

```
adult <- read.csv(  
  "data/adult.data",  
  header = FALSE,  
  strip.white = TRUE,  
  na.strings = "?"  
)  
  
class(adult)
```

```
[1] "data.frame"
```

```
dim(adult)
```

```
[1] 32561    15
```

b.

```
names(adult) <- c(  
  "age", "workclass", "fnlwgt", "education", "education_num",  
  "marital_status", "occupation", "relationship", "race", "sex",  
  "capital_gain", "capital_loss", "hours_per_week",  
  "native_country", "income"  
)  
  
names(adult)
```

```
[1] "age"          "workclass"    "fnlwgt"      "education"
[5] "education_num" "marital_status" "occupation"   "relationship"
[9] "race"         "sex"          "capital_gain" "capital_loss"
[13] "hours_per_week" "native_country" "income"
```

c.

```
colSums(is.na(adult))
```

age	workclass	fnlwgt	education	education_num
0	1836	0	0	0
marital_status	occupation	relationship	race	sex
0	1843	0	0	0
capital_gain	capital_loss	hours_per_week	native_country	income
0	0	0	583	0

Missing values occur only in `workclass`, `occupation`, and `native_country`. There are 1836 missing observations for `workclass`, 1843 for `occupation`, and 583 for `native_country`. All other variables have no missing observations.

d.

```
nrow(adult)
```

```
[1] 32561
```

```
adult_filtered <- adult[adult$hours_per_week >= 10, ]
```

```
nrow(adult_filtered)
```

```
[1] 32103
```

The sample size is 32,561 before filtering and 32,103 after restricting the sample to individuals working at least 10 hours per week.

e.

```
nrow(adult_filtered)
```

```
[1] 32103
```

```
adult_complete <- adult_filtered[
  !is.na(adult_filtered$occupation) &
  !is.na(adult_filtered$workclass) &
  !is.na(adult_filtered$native_country),
]

nrow(adult_complete)
```

```
[1] 29873
```

```
colSums(
  is.na(
    adult_complete[,c("occupation", "workclass", "native_country")]
  )
)
```

occupation	workclass	native_country
0	0	0

Observations with missing values in occupation, workclass, or native_country were removed because these variables are needed for the analysis, and observations with missing values in these fields would have incomplete information.

f.

```
adult_clean <- adult_complete[
  adult_complete$age >= 18 &
  adult_complete$age <= 90 &
  adult_complete$education != "Preschool",
]

range(adult_clean$age)
```

```
[1] 18 90
```

```
sum(adult_clean$education == "Preschool")
```

```
[1] 0
```

g.

```
nrow(adult_clean)
```

```
[1] 29523
```

The final sample size is 29,523 observations.